Advanced Statistical Physics

Lecture Notes

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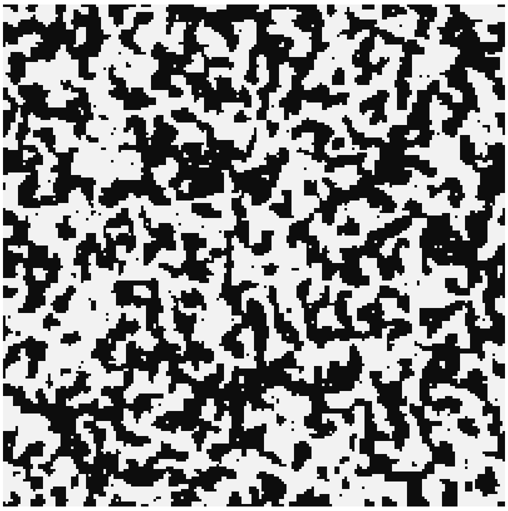

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Lecture 15: Landau-Ginzburg Theory

In the Landau theory, we had assumed an average value m of the magnetisation for the entire system and used this as the order parameter. However, this method is too rough as it misses any spatial dependence of the order parameter. m was spatially homogenous which perhaps is not what we want. We now generalise $m \to m(\mathbf{x})$ where $m(\mathbf{x})$ is now a field.

We focus on magnetic systems only, however, this can be applied to more general systems. To see this clearly, divide the system into N' different cells $\{I_1, I_2 \dots I_{N'}\}$, each with a characteristic length $\sim l$ as shown below. Here l is a mesoscopic length scale, $a \ll l \ll L$ where a is the lattice spacing and L is the typical linear dimension of the lattice.

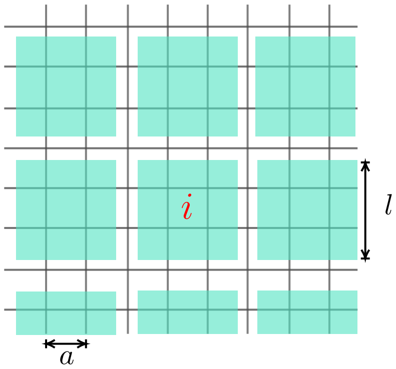


Figure 1: Coarse graining the lattice

Let the centre each cell I be $\mathbf{x}$ and define the local order parameter,

$$m_{\mathrm{I}}(\mathbf{x}) = \frac{1}{N_{\mathrm{I}}} \sum_{i \in I} \sigma_i \tag{1}$$

where N_{I} is the number of spins in cell I. This process is called *coarse graining*, since we are defining an order parameter, averaged over N_{I} spins in each cell. Let us now define $\mathcal{L}$ in this case. At the first glance, it seems that

$$\mathcal{L}(\{m_{\mathrm{I}}\}) = \sum_{i}^{N'} \left[\frac{1}{2} a_2 (T - T_c) m_i^2 + \frac{1}{4} a_4 m_i^4 - h_i m_i \right]$$

We have suppressed the $\mathbf{x}$ parameter, since $\mathbf{x}$ is dependent on i itself. Note that the above free energy does not take into account the cost of maintaining the local magnetisation in adjacent cells. Every m_i is independent and this can lead to large difference in the order parameter in adjacent cells. We assume that the spatial variation of the local order parameter is 'slow' and for this, we introduce a symmetric coupling term,

$$\mathcal{L}(\{m_{\mathrm{I}}\}) = \sum_{i}^{N'} \left[\frac{1}{2} a_2 (T - T_c) m_i^2 + \frac{1}{4} a_4 m_i^4 - h_i m_i \right] + \frac{1}{2} \sum_{ij} \mathcal{C}_{ij} (m_i - m_j)^2 \tag{2}$$

where $\mathcal{C}_{ij}$ is the short-range coupling and based on a characteristic cutoff length R, is defined as,

$$\mathcal{C}_{ij} = \begin{cases} c_0 & |i - j| l < R \\ 0 & |i - j| l > R \end{cases}$$

There are some subtleties here though. Note that $\mathbf{x}$ can take only discrete values, based on the position of the cell. Hence we assume that $\frac{N'}{N}$ is small enough that $\mathbf{x}$ can be taken to be continuous. Moreover, within each cell, $m_{\mathrm{I}}(\mathbf{x})$ is quantised in terms of $\frac{1}{N_{\mathrm{I}}}$. We choose N_{I} large enough, so that $m_{\mathrm{I}}(\mathbf{x})$ can take any value between -1 and $+1$. This allows us to replace $m_{\mathrm{I}}(\mathbf{x})$ by just a continuous function $m(\mathbf{x})$. Let us now manipulate the coupling term now. Assume a 1D system for ease.

$$\begin{aligned}\frac{1}{2}\sum_{ij}\mathcal{C}_{ij}(m_i-m_j)^2 &= \sum_i\sum_{j=1}^{R/l} c_0(m_{i+j}-m_i)^2\\ &= \sum_i\sum_{j=1}^{R/l} c_0\left[\frac{(m_{i+j}-m_i)^2}{(jl)^2}\right](jl)^2\end{aligned} \tag{3}$$

The thing in the bracket looks like the discrete version of a derivative. In the continuum limit, this leads to $(\partial_x m(x))^2$, which becomes the gradient term $(\boldsymbol{\nabla} m(\mathbf{x}))\cdot(\boldsymbol{\nabla} m(\mathbf{x}))$ in general d dimensions. The sum over i can be converted to a d-dimensional integral,

$$\sum_i \equiv \frac{1}{l^d}\int \mathrm{d}^d x$$

where l^d is the elemental volume in each cell with centre $\mathbf{x}$. The sum over j is just a constant and depends on the interaction range R. Thus, absorbing every constants into a single parameter γ, the coupling term becomes,

$$\frac{1}{2}\sum_{ij}\mathcal{C}_{ij}(m_i-m_j)^2 \simeq \frac{\gamma}{2}\int \mathrm{d}^d x\ (\boldsymbol{\nabla} m(\mathbf{x}))^2$$

The final expression for the Landau free energy is,

$$\mathcal{L}[m(\mathbf{x})] = \int \mathrm{d}^d x\ \left[\frac{1}{2}a_2(T-T_c)(m(\mathbf{x}))^2 + \frac{1}{4}a_4(m(\mathbf{x}))^4 - h(\mathbf{x})m(\mathbf{x}) + \frac{\gamma}{2}(\boldsymbol{\nabla} m(\mathbf{x}))^2\right] \tag{4}$$

Note that, $\mathcal{L}$ has now become a *functional*, that is, it takes a function $m(\mathbf{x})$ and spits out a scalar. The partition function calculation becomes a bit involved, since it contains the term $\exp(-\beta\mathcal{L}[m(\mathbf{x})])$ and we have to perform an integral. However, this is not an ordinary integral, since the integration variable is now a function $m(\mathbf{x})$. What happens is that there are infinite realisations of the function $m(\mathbf{x})$ and we need to 'integrate' over all these realisations. The process through which it is done is called *functional integral* or a more cool sounding *path integral.* The formal notation of the functional integral is,

$$\mathcal{Z} = \int \mathcal{D}m(\mathbf{x})\ e^{-\beta\int \mathrm{d}^d x\ \left[\frac{1}{2}a_2(T-T_c)(m(\mathbf{x}))^2+\frac{1}{4}a_4(m(\mathbf{x}))^4-h(\mathbf{x})m(\mathbf{x})+\frac{\gamma}{2}(\boldsymbol{\nabla} m(\mathbf{x}))^2\right]} \tag{5}$$

Out of the infinite realisations of $m((x))$, the one which gives the equation of state is the one which minimises the free energy functional. For minimisation purpose, we have to invoke the concept of function derivative since $m(\mathbf{x})$ is a function. Thus, we have the equation of state as,

$$m^*(\mathbf{x})\ :\quad \frac{\delta\mathcal{L}}{\delta m} = 0$$

Before formally going down this rabbit hole, let us find the minima by simply calculating the variation $\delta\mathcal{L}$ when the function $m(vbx)$ is varied from the minimising function as $m((x)) = m^*(\mathbf{x}) + \delta m(\mathbf{x})$, such that at the boundaries, $\delta m = 0$.

$$\delta\mathcal{L} = \int \mathrm{d}^d x\ \left[a_2(T-T_c)m\delta m + a_4 m^3\delta m - h\delta m + \frac{\gamma}{2}\times 2(\delta(\boldsymbol{\nabla} m)\cdot(\boldsymbol{\nabla} m))\right]$$

Let us calculate the variation of the gradient term.

$$\begin{aligned}\int \mathrm{d}^d x\ (\boldsymbol{\nabla} m)\delta(\boldsymbol{\nabla} m) &= \int \mathrm{d}^d x\ (\boldsymbol{\nabla} m)\boldsymbol{\nabla}(\delta m)\\ &\overset{\text{IBP}}{=} -\int \mathrm{d}^d x\ (\boldsymbol{\nabla}^2 m)\delta m + [\boldsymbol{\nabla} m\delta m]_{\text{bound.}}\\ &= -\int \mathrm{d}^d x\ (\boldsymbol{\nabla}^2 m)\delta m\end{aligned}$$

The boundary term is zero, since δm is zero at the boundaries. From the variation of the functional, we obtained

$$\delta\mathcal{L} = \int \mathrm{d}^d x \; \left[a_2(T-T_c)m(\mathbf{x}) + a_4(m(\mathbf{x}))^3 - h(\mathbf{x}) - \gamma\boldsymbol{\nabla}^2 m(\mathbf{x})\right]\delta m = 0$$

Since δm is arbitrary, we have the equation of state satisfied by,

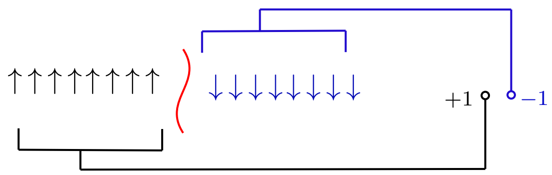
$$\boxed{a_2(T-T_c)m^*(\mathbf{x}) + a_4(m^*(\mathbf{x}))^3 - h(\mathbf{x}) - \gamma\boldsymbol{\nabla}^2 m^*(\mathbf{x}) = 0}$$

Lecture 16: Kink Solutions

In the previous lecture, we introduced a spatial variation of the order parameter, $m(\mathbf{x})$. Consider the 1D Ising chain as an example. Suppose in the extreme left end of the chain, we have a $\uparrow$ spin while in the extreme right, we have a $\downarrow$ spin.

If we keep this constraint, then it is evident that somewhere in the middle, there should be a spin flip (that is, we cannot *continuously deform* the left side to bring it to the right side). This inability to continuously deforma aptly indicates that this is a *topological artifact*.

This spin flip is nothing but a *domain wall* if we think about it. The domain wall is also exotically called a *kink* or *soliton* depending on the context and the type of system used. Note that if there are two kinks, then there are two spin flips and hence these cancel each other (that is, the boundaries have the same spins). Thus, this configuration can be continuously deformed to the ground state.



The spins left and right of the kink can be mapped from the real space (of the lattice) to the order-parameter space, where there are two points $+1$ and -1, as shown above. Thus a state $(+,+)$ or $(-,-)$ will have all up spins and down spins respectively and hence are the ground states. The state $(+,-)$ (state (-,+)) will have up spins on the left (right) and down spins on the right (left) of the kink. These are the four topologically distinct sectors for the 1D Ising model.

We want to see the variation of the order parameter across the domain wall. For this, consider the following Landau functional,

$$\mathcal{L}[m(\mathbf{x})] = \int \mathrm{d}^d x \; \left[\frac{1}{2}a_2(T-T_c)(m(\mathbf{x}))^2 + \frac{1}{4}a_4(m(\mathbf{x}))^4 - h(\mathbf{x})m(\mathbf{x}) + \frac{\gamma}{2}(\boldsymbol{\nabla} m(\mathbf{x}))^2\right]$$

Now, suppose the in only one specific direction, there is spatial variation while all the other $(d-1)$ dimensions are spatially uniform. Hence $m(\mathbf{x}) \equiv m(x)$ and let A be the integral over the rest $(d-1)$ dimensions.

$$\frac{\mathcal{L}}{A} = \int \mathrm{d}x \; \left[\frac{1}{2}a_2(T-T_c)(m(x))^2 + \frac{1}{4}a_4(m(x))^4 - h(x)m(x) + \frac{\gamma}{2}\left(\frac{\partial m(x)}{\partial x}\right)^2\right] \tag{6}$$

Considering $h(x) = 0$, we get the equation of motion:

$$\gamma\frac{\mathrm{d}^2 m}{\mathrm{d}x^2} = a_2(T-T_c)m + a_4 m^3 \tag{7}$$

Multiplying both sides by $\frac{\mathrm{d}m}{\mathrm{d}x}$ we get,

$$\begin{aligned}\gamma\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)\frac{\mathrm{d}^2m}{\mathrm{d}x^2} = \left[a_2(T-T_c)m + a_4m^3\right]\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right) &\implies \frac{\gamma}{2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)^2 = \frac{\partial}{\partial x}\left[\frac{1}{2}a_2(T-T_c)m^2 + \frac{1}{4}a_4m^4\right] \\ &\implies \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\gamma}{2}\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)^2 - \frac{1}{2}a_2(T-T_c)m^2 - \frac{1}{4}a_4m^4\right] = 0 \\ &\implies -\frac{\gamma}{2}\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)^2 + \frac{1}{2}a_2(T-T_c)m^2 + \frac{1}{4}a_4m^4 = C\end{aligned}$$

where C is a constant determined by the boundary conditions. So let's set $m(x \to +\infty) = m_\infty = \sqrt{-\frac{a_2(T-T_c)}{a_4}}$ and $\frac{\mathrm{d}m}{\mathrm{d}x}\Big|_\infty = 0$. Using this, we obtain the value of C

$$C = \frac{1}{2}a_2(T-T_c) \times \left(-\frac{a_2(T-T_c)}{a_4}\right) + \frac{1}{4}a_4 \times \frac{(a_2(T-T_c))^2}{a_4^2} = -\frac{[a_2(T-T_c)]^2}{4a_4}$$

Substituting this value above, we get the equation as,

$$\begin{aligned}0 &= -\frac{\gamma}{2}\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)^2 + \frac{1}{2}a_2(T-T_c)m^2 + \frac{1}{4}a_4m^4 + \frac{[a_2(T-T_c)]^2}{4a_4} \\ &= -\frac{\gamma}{2}\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)^2 + \frac{a_4}{4}\left[\frac{2a_2}{a_4}(T-T_c)m^2 + m^4 + m_\infty^4\right] \\ &= -\frac{\gamma}{2}\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)^2 + \frac{a_4}{4}\left[-2m^2m_\infty^2 + m^4 + m_\infty^4\right] \\ &= -\frac{\gamma}{2}\left(\frac{\mathrm{d}m}{\mathrm{d}x}\right)^2 + \frac{a_4}{4}\left(m^2 - m_\infty^2\right)^2\end{aligned} \tag{8}$$

from which get two solutions,

$$\sqrt{\frac{2\gamma}{a_4}}\frac{\mathrm{d}m}{\mathrm{d}x} = \pm(m^2 - m_\infty^2)$$

Appendices

From the first law we have the energy conservation equation,

$$\mathrm{d}U = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N$$

and thus the internal energy is a function of S, V and N. We know that the internal energy is a homogenous function of degree 1, on physical grounds, since if we scale the system variables by λ, U is also appropriately scaled (extensitivity)

$$U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$$

Now, from Euler's theorem of homogenous functions, we know that any homogenous function f of degree k can be written as,

$$kf(x_1, x_2 \dots x_n) = \sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i}$$

Using this theorem for U, we have

$$U = S\frac{\partial U}{\partial S} + V\frac{\partial U}{\partial V} + N\frac{\partial U}{\partial N}$$

Substituting the values from the first law, we obtain $U = TS - PV + \mu N$

A. Legendre Transformation

From classical mechanics, we know that the Legendre transformation is used to convert Lagrangian $L(x, \dot{x})$ to the Hamiltonian $H(x, p)$ and vice versa. Note how we switch $\dot{x}$ in L for p in H. We will call the variable that we want to switch to as the 'conjugate' of the variable we are switching. The Legendre transformation of the Lagrangian gives us the Hamiltonian,

$$H(x, p) = p\dot{x} - L$$

A function $f(x)$ with the Legendre transform $f^*(p)$ satsifies $f(x) + f^*(p) = xp$. We will use this for thermodynamic quantities also, however we will use an alternate definition using minus sign, $f(x) - f^*(p) = xp$. From the first law, we know that,

$$U = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N$$

As clearly seen, the conjugate variable pairs are (S, T), $(V, -P)$, (N, μ). We will consider the starting point as the internal energy $U(S, V, N)$ which is a function of S, V, N from the statement of the first law. However, in any experiment, keeping these as control parameters are way too hard. Think about this, controlling entropy of a system is an arduous task but controlling its conjugate variable, T, is easy-peasy.

We get four different 'potentials' by just Legendre transformation (which is basically multiplying the conjugate variables together and subtracting from the function) of the internal energy, switching between the different conjugate variables.

- **Helmholtz Free Energy**: switching $S \leftrightarrow T$

$$F(T, V, N) = U - TS \implies \mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V + \mu\mathrm{d}N$$

- **Enthalpy**: switching $V \leftrightarrow -P$

$$H(S, P, N) = U + PV \implies \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Gibb's Free Energy**: switching both $S \leftrightarrow T$ and $V \leftrightarrow -P$

$$G(T, P, N) = U - TS + PV \implies \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Grand Potential**: switching both $S \leftrightarrow T$ and $N \leftrightarrow \mu$

$$\Phi(T, V, \mu) = U - TS - \mu N = -PV \implies \mathrm{d}\Phi = -S\mathrm{d}T - P\mathrm{d}V - N\mathrm{d}\mu$$

B. 1D Solution of Ising Model

In 1D, the Ising model is given by,

$$\mathcal{H} = -J\sum_i^N \sigma_i\sigma_{i+1} - h\sum_i \sigma_i$$

where we have assumed periodic boundary conditions. The second term can be written as a symmetric sum, utilisng the translation invariance. Hence the Hamiltonian becomes,

$$\mathcal{H} = -J\sum_i^N \sigma_i\sigma_{i+1} - \frac{h}{2}\sum_i (\sigma_i + \sigma_{i+1}) \equiv \sum_i \mathcal{E}(\sigma_i, \sigma_{i+1})$$

where $\mathcal{E}(\sigma_i, \sigma_{i+1}) = -J\sigma_i\sigma_{i+1} - \frac{B}{2}(\sigma_i + \sigma_{i+1})$. Note that $\mathcal{E}(+,+) = -(J+h), \mathcal{E}(-,-) = -(J-h), \mathcal{E}(+,-) = \mathcal{E}(-,+) = J$. The contribution of each configuration is,

$$P(\{\sigma_i\}) \propto e^{-\beta\mathcal{H}(\{\sigma_i\})} = \exp\left(-\beta\sum_j \mathcal{E}(\sigma_j, \sigma_{j+1})\right) \equiv \prod_{i=1}^N \exp(-\beta\mathcal{E}(\sigma_i, \sigma_{i+1}))$$

Let us define $t_{\sigma_i\sigma_{i+1}} = \exp(-\beta\mathcal{E}(\sigma_i, \sigma_{i+1}))$ which is an entry of a matrix t called the *transfer matrix*:

$$t_{\sigma_i\sigma_j} \equiv \langle\sigma_i|t|\sigma_j\rangle$$

The transfer matrix is just a notational trick and is assumed to be an operator of the Hilbert space spanned by $|{+1}\rangle$ and $|{-1}\rangle$.

$$\begin{aligned} \langle+|t|+\rangle &= e^{\beta(J+h)} \qquad & \langle+|t|-\rangle &= e^{-\beta J} \\ \langle-|t|-\rangle &= e^{\beta(J-h)} \qquad & \langle-|t|+\rangle &= e^{-\beta J} \end{aligned}$$

The partition function can then be written as,

$$\begin{aligned} \mathcal{Z} = \sum_{\{\sigma_i\}} e^{-\beta\mathcal{H}(\{\sigma_i\})} &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} t_{\sigma_1\sigma_2}t_{\sigma_2\sigma_3}\ldots t_{\sigma_N\sigma_1} \\ &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} \langle\sigma_1|t|\sigma_2\rangle\langle\sigma_2|t|\sigma_3\rangle\langle\sigma_3|t|\sigma_4\rangle\ldots\langle\sigma_N|t|\sigma_1\rangle \end{aligned} \tag{9}$$

From the resolution of identity, $\sum\limits_{\sigma}|\sigma\rangle\langle\sigma| = \mathbb{1}$ and hence all the inner spin sums vanish. We are left with,

$$\mathcal{Z} = \sum_{\sigma_1}\langle\sigma_1|t^N|\sigma_1\rangle = \mathrm{Tr}(t^N) = \lambda_+^N + \lambda_-^N = \lambda_+^N\left(1 + \left(\frac{\lambda_-}{\lambda_+}\right)^N\right)$$

where $\lambda_- < \lambda_+$ are the eigenvalues of the transfer matrix. In the limit $N\to\infty$, $\mathcal{Z}\to\lambda_+$. The eigenvalues of the transfer matrix are given as,

$$t \equiv \begin{pmatrix} e^{\beta(J+h)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-h)} \end{pmatrix} \implies \boxed{\lambda_\pm = e^{\beta J}\left[\cosh(\beta h) \pm \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right]}$$

In the thermodynamic limit, the free energy $F = -k_{\mathrm{B}}T\ln\mathcal{Z} = -Nk_{\mathrm{B}}T\ln\lambda_+$ and then the magnetisation is given as,

$$\begin{aligned} M = -\frac{\partial F}{\partial h} &= Nk_{\mathrm{B}}T\frac{\partial}{\partial h}\left[\beta J + \ln\left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)\right] \\ &= Nk_{\mathrm{B}}T\frac{\beta\sinh(\beta h) + \frac{2\beta\sinh(\beta h)\cosh(\beta h)}{2\sqrt{\sinh^2(\beta h)+e^{-4\beta J}}}}{\left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)} \\ &= N\cancel{(k_{\mathrm{B}}T)}\cancel{\beta}\frac{\sinh(\beta h)\left[\sqrt{\sinh^2(\beta h) + e^{-4\beta J}} + \cosh(\beta h)\right]}{\left(\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)\left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right)} \\ &= \frac{N\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}} \end{aligned}$$

The magnetisation per spin is then,

$$\boxed{m = \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}}$$

For $T > 0$ and $h \neq 0$, m is a well-behaved function with no singularity. For $h \to 0$, we get $m \to 0$ for all $T > 0$. Thus, temperature is not sufficient to produce a finite magnetisation without an external field. For $T \to 0$, we have $m \to 1$ and hence we can say that the phase transition to ferromagnetic behaviour in the 1D Ising model is at $T_c = 0$. This is the reason why Ising said that there is no phase transition in 1D case (and incorrectly generalised it to higher dimensions too)!

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